

Project Proposal DS 3002 Data Mining

Surgical Data Analysis

Group Members:

Sohaib Ahmed (21L-5635)

Muhammad Hassan Khalid (21L-5692)

Muhammad Hamza (21L-5636)



Problem and Dataset Description:

The objective of this project is to develop predictive models for binary classification, specifically to predict the likelihood of surgical complications based on patient characteristics and surgical parameters. The dataset utilized for this analysis is sourced from Kaggle

(<https://www.kaggle.com/datasets/omnamahshivai/surgical-dataset-binary-classification/data>). It encompasses comprehensive information pertaining to surgical procedures and patient details, aiming to forecast surgical complications. The dataset comprises features including patient demographics, medical history, and surgical parameters. Our task involves constructing predictive models capable of accurately forecasting the likelihood of complications during surgical procedures based on these features.

Pre-Processing Techniques:

1. Handling Missing Values: Identify and address missing values in the dataset to ensure completeness and accuracy.
2. Outlier Detection and Removal: Remove outliers to ensure the robustness and reliability of the predictive models.
3. Feature Scaling: Standardize numerical features to bring them to a similar scale, preventing features with larger magnitudes from dominating the model training process.
4. One-Hot Encoding: Encode categorical variables using one-hot encoding to convert them into a numerical format suitable for machine learning algorithms.

5. Dimensionality Reduction: Implement techniques such as Principal Component Analysis (PCA) to reduce the dimensionality of the dataset while preserving important information, aiding in visualization and model performance.
6. Feature Selection: Select relevant features based on correlation analysis or domain knowledge to improve model interpretability and performance.
7. Data Imputation: Fill missing values using appropriate methods such as mean, median, or mode imputation to maintain the integrity of the dataset.
8. Column Transformation: Transform columns using techniques like `ColumnTransformer()` to apply specific preprocessing steps to different types of features (e.g., numerical vs. categorical) efficiently.

Models/Algorithms/Techniques:

1. Logistic Regression
2. Random Forest Classifier
3. Gradient Boosting Classifier
4. Support Vector Machines (SVM)
5. K-Nearest Neighbors (KNN)
6. Decision Trees
7. Naive Bayes
8. Neural Networks

Implementation Approach for Models:

1. Load and preprocess the dataset
2. Split the dataset into training and testing sets.
3. Implement various machine learning and neural network models using libraries such as scikit-learn and TensorFlow.
4. Train the models on the training data.
5. Evaluate the models using metrics such as accuracy, precision, recall, and F1-score.
6. Select the best-performing model based on evaluation metrics and feature importance analysis.

Software Tools:

1. Python programming language
2. Jupyter Notebook for code development and documentation
3. Libraries such as pandas, numpy, matplotlib, seaborn, scikit-learn, and TensorFlow for data manipulation, visualization, and modeling.

Expected Results and Evaluation:

1. Accurate prediction of surgical complications based on patient characteristics and surgical parameters.
2. Evaluation metrics including accuracy, precision, recall, and F1-score to measure model performance.
3. Identification of key features contributing to surgical complications through feature importance analysis.

Preliminary Result and Dataset Explored:

1. Initial exploratory data analysis to understand the structure and content of the dataset.
2. Preprocessing steps such as handling missing values, outlier detection, and feature scaling.
3. Implementation of baseline machine learning models for initial evaluation.

Outline of the work-to-do:

1. Complete preprocessing steps including handling missing values, outlier detection, and feature scaling.
2. Implement and evaluate multiple machine learning models and neural network model using the preprocessed data.
3. Select the best-performing model based on evaluation metrics.
4. Perform feature importance analysis to identify key predictors of surgical complications.

5. Generate insights and recommendations based on the analysis for medical practitioners and researchers.

Additional Points:

1. Employ model evaluation techniques including cross-validation to assess model generalization performance.
2. Calculate error metrics including Mean Squared Error (MSE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) for comprehensive model evaluation.
3. Combine results from different feature sets and cross-validation to identify the best-performing model.

Visualization and Interpretation:

Utilize various visualization techniques including ROC curves, training/validation loss over epochs, and bar charts comparing model performance. Interpret the findings to provide actionable insights for medical practitioners and researchers.